

Haider Ali

LLM / AI Engineer • RAG • Agentic AI • Computer Vision

Karachi, Pakistan

PROFILE

Computer Science undergraduate with roughly two years of self-directed, hands-on AI/ML development spanning LLMs, RAG, computer vision, and agentic systems. Comfortable picking up new tools and diving into unfamiliar problem spaces quickly. Built systems including a 7-agent autonomous deal-finder, a hybrid-retrieval RAG assistant, and LLMs fine-tuned with LoRA/PEFT. Seeking an LLM/AI Engineer internship or part-time role to bring that build-first mindset to a real team.

EDUCATION

Bachelor of Science in Computer Science | Federal Urdu University of Arts, Science & Technology, Karachi | Jan 2025 – Dec 2028

Focus areas: machine learning, deep learning, natural language processing, computer vision, large language models, RAG, LLM fine-tuning, and agentic AI.

TECHNICAL SKILLS

Languages: Python, C/C++

AI/ML: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

Generative AI: LLMs, RAG, Embeddings, LoRA/PEFT, Prompt Engineering, Agentic AI

Frameworks: LangChain, Hugging Face Transformers, Flask

Vector/Retrieval: ChromaDB, BM25, Cross-Encoder Reranking

Computer Vision: OpenCV, MediaPipe

Databases & Web Scraping: PostgreSQL, BeautifulSoup, Selenium

UI: Gradio, pywebview, PyQt5, PySide6, HTML, CSS, JavaScript (LLM-assisted frontend development)

Tools: Git, GitHub, Jupyter, VS Code

SELECTED PROJECTS

DealScout / Autonomous AI Deal Finder | *Python, LLM Agents, RAG, PyTorch*

- **Architected a 7-agent pipeline** (Scanner, Planning/Autonomous Planning, Ensemble, Specialist, Frontier, Neural Network, Messaging) that scans RSS deal feeds and autonomously identifies underpriced products
- Combined **3 independent pricing signals** into an ensemble estimate: a Llama 3.2 3B model fine-tuned with QLoRA and deployed on Modal (T4 GPU), a RAG estimator retrieving the 5 nearest comparable products via ChromaDB, and a custom PyTorch neural network
- Built both a fixed workflow and a fully autonomous mode where the LLM dynamically selects which agents/tools to call; shipped as a Gradio app with live agent-log streaming and 3D embedding visualization
- Auto-notifies via ntfy when the estimated discount exceeds a **\$50 threshold**, using an LLM to draft the alert

Fifa_Box / Hybrid RAG System | *Python, LangChain, ChromaDB, BM25*

- Built a hybrid RAG assistant spanning **93 years of FIFA World Cup history (1930–2022)**, combining dense vector retrieval with BM25 keyword search, query rewriting, and cross-encoder reranking
- Built a custom evaluation suite (**179 test cases**) scoring retrieval quality (MRR, nDCG, **Precision@5 ≈ 0.80**) and answer quality via LLM-as-judge (accuracy, completeness, relevance, faithfulness)
- Shipped as a desktop app (pywebview) with a hand-built HTML/CSS/JS frontend, single in-process app, no backend server

AI Brochure Generator | *GPT-4o-mini, Gemini 2.0 Flash, Gradio, BeautifulSoup*

- Built an AI brochure generator that scrapes a company site, uses an LLM to auto-select relevant subpages (About, Careers, etc.), then generates a full brochure via GPT-4o-mini or Gemini 2.0 Flash, streamed token-by-token in a Gradio UI

Shadow_Clone_Jujutsu / Real-Time Hand Gesture Recognition | *MediaPipe, OpenCV, TensorFlow, Scikit-learn*

- Collected and labeled a **~2,000-sample** hand-gesture dataset; trained a custom TFLite classifier on MediaPipe hand landmarks for real-time recognition
- Paired it with a person-segmentation model to trigger webcam-driven visual and audio effects live

LANGUAGES

Urdu (Native) | English (Fluent)